ASSOCIATE PROFESSOR
SCRIPPS INSTITUTION OF OCEANOGRAPHY
UNIVERSITY OF CALIFORNIA SAN DIEGO
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# Alice-Agnes Gabriel

I am a computational earthquake physicist combining simulations, routinely using the largest supercomputers worldwide, with data-driven techniques and theoretical analysis to tackle some of the grand challenges of Geophysics. My group's research bridges space-time scales and disciplines across Seismology, Tsunami Science, Computer Science, and Applied Mathematics. We integrate interdisciplinary Earth observations with novel modeling approaches to study a range of multi-physics problems.

## Education

2022 **Habilitation** (Dr. habil.), *Geophysics*, Ludwig-Maximilians-Universität (LMU) München, Munich, Germany.
Cumulative habilitation thesis "Computational Earthquake Physics (Computergestützte Erdbebenphysik)"

2013 **Dr. Sci.** (PhD), *Geophysics*, Swiss Seismological Service, ETH Zurich, Switzerland.
Thesis "Physics of dynamic rupture pulses and macroscopic earthquake source properties in elastic and plastic media", Diss. ETH No. 20567
Advisors: J.-P. Ampuero (CALTECH), P.M. Mai (KAUST), L.-A. Dalguer & D. Giardini (ETH)

2008 **Diplom** (MSc), *Physics*, Technical University Dresden, Germany.
Majors: Theoretical physics, Solid-state physics, Biophysics
Thesis "Atomistic Simulations of Solid-phase Epitaxial Regrowth of Amorphous Germanium", supervisors M. Posselt, W. Möller (Helmholtz Center Dresden-Rossendorf)

2006 **Certificate of Proficiency**, *Geophysics, Geology*, University of Auckland, New Zealand.

2004 **Vordiplom** (BSc), *Physics*, Technical University Dresden, Germany.

## Professional Appointments

2022– **Associate Professor**, *Geophysics*, Scripps Institution of Oceanography, UC San Diego, La Jolla, CA.

2021–2022 **Assistant Professor**, *Geophysics*, Scripps Institution of Oceanography, UC San Diego, La Jolla, CA.
(on leave without pay 11/2021–06/2022)

2022– **Guest Professor** (2022–24) **& Affiliated Researcher** (2024–), LMU Munich, Germany.

2021–2022 **Professor** (W2), *Earthquake Physics*, LMU Munich, Germany.

2014–2021 **Assistant Professor** (akadem. Rat a.Z.), *Seismology/Geophysics*, LMU Munich, Germany.
(26 months maternity leave: 10/2014–02/2016 and 05/2016–02/2017)

2012–2014 **Postdoctoral Researcher**, *Seismology/Geophysics*, LMU Munich, Germany.

2008–2012 **Research Assistant**, *Swiss Seismological Service*, ETH Zurich, Switzerland.

2007–2008 **Research Assistant**, Helmholtz Center Dresden-Rossendorf (HZDR), Dresden, Germany.

2005–2006 **Student Assistant** (part time), AMD Advanced Mask Technology Center, Dresden, Germany.

2003–2005 **Student Assistant** (part time), Leibniz Institute for Solid State and Materials Research, Dresden, Germany.

### Visiting Appointments

2018/19 **Visiting Professor**, King Abdullah University of Science and Technology (KAUST), Saudi Arabia.

## Awards and Honors

2026 **Top Cited Article Award** (AGU/Wiley), Journal of Geophysical Research: Solid Earth, for one student-led paper: Palgunadi et al. (2024, [66]), among the ten most-cited JGR: Solid Earth articles published in 2024

2026 **3 Top Viewed Article Awards** (AGU/Wiley) for three student-led papers: Wong et al., JGR: Solid Earth (2024, [72]); Kutschera et al., Geophysical Research Letters (2024, [75]); Palgunadi et al., JGR: Solid Earth (2024, [66])

2025 **ACM Gordon Bell Prize**, Association for Computing Machinery, awarded at the International Conference for High Performance Computing, Networking, Storage, and Analysis (SC25), St. Louis, USA, for [98] "Real-time Bayesian inference at extreme scale: A digital twin for tsunami early warning applied to the Cascadia subduction zone". Extremely competitive, annually awarded prize for outstanding achievements in high-performance computing

2025 **HPC Innovation Excellence Award**, Hyperion Research, presented at SC25 for the Cascadia earthquake-tsunami digital twin [98]

2025 **HPCwire Readers' Choice Award**, Best Use of HPC in Physical Sciences, presented at SC25, also for [98]

2025 **Winner, MFEM Community Workshop Visualization Contest**, for the visualization by S. Henneking et al., of the Cascadia earthquake-tsunami digital twin (2025, [98])

2025 **S.N. Jepson Lecturer**, Victoria University of Wellington, New Zealand. Named distinguished lectureship

2024–2026 **Thomas and Evelyn Page Chancellor's Endowed Faculty Fellowship in Geosciences**, UC San Diego

2023 **James B. Macelwane Medal**, American Geophysical Union (AGU), awarded in recognition of significant contributions to the geophysical sciences by an outstanding early-career scientist. Conferred with election as AGU Fellow

2022 **C.F. Gauss Lecturer**, German Geophysical Society

2022 **EGU Elected President Seismology Division**

2022 **Humboldt Foundation, Japanese-American-German Frontiers Science Symposium Invited Participant**

2020 **Charles F. Richter Early Career Award** of the Seismological Society of America. SSA award for innovative research in earthquake rupture dynamics and tsunami genesis, among other topics, using physics-based models and high-performance computing

2020 **PRACE Ada Lovelace Award for HPC**. Annual award for a female scientist making an outstanding contribution and impact on High Performance Computing (HPC) in Europe and the world and serves as a role model for women

2019 **ERC Starting Grant Awardee**. Highly competitive research grant by the European Research Council. The evaluation is based on scientific excellence and the PI's high potential for research independence

2019 **Appointment to the LMU Center of Advanced Studies – Young Center**

2019 **Top Downloaded JGR Paper**, van Zelst et al., 2019, was recognized as one of the most read in J. Geophys. Res.-Solid Earth

2019 **Top 10 Altmetric GJI Article of 2018**. Wollherr et al., 2018, was one of the top 10 articles published in Geophysical Journal International with the highest Altmetric scores and as such made freely available through 2019

2018/19 **Visiting Professor Scholarship**, KAUST King Abdullah University of Science and Technology, Saudi Arabia. Teaching of short-courses "Dynamic Rupture on 3D complex fault systems" and "SeisSol: An ADER-DG code for fracture dynamics". Collaboration on three-dimensional rupture dynamics across complex fault systems for the Gulf of Aqaba fault system and cascading fractures in geo-reservoirs

2018 **SC18 Selected Paper for Student Cluster Competition Reproducibility Challenge**, IEEE/ACM International Conference for High Performance Computing, Networking, Storage, and Analysis (SC18), Dallas, USA. 16 teams of undergraduate students reproduced our SC17 awarded paper (Uphoff et al., 2017) in a real-time, non-stop, 48-hour challenge

2017 **SC17 Best Paper Award**, IEEE/ACM International Conference for High Performance Computing, Networking, Storage, and Analysis (SC17), Denver, USA. Most prestigious prize for application driven high-performance computing awarding the largest multi-physics simulations to date modelling the 2004 Sumatra-Andaman earthquake

2014 **ACM Gordon Bell Prize Finalist**, Association for Computing Machinery (ACM) at International Conference for High Performance Computing, Networking, Storage, and Analysis (SC14), New Orleans, USA. Highly competitive, annually awarded prize for outstanding achievements in high-performance computing. In our submission, we performed multi-physics simulations of a 1992 Landers earthquake scenario on the largest existing supercomputers

2014 **PRACE ISC Award**, Partnership for Advanced Computing in Europe (PRACE) at the International Supercomputing Conference ISC14, Leipzig, Germany. Awarded for producing the first simulations that obtained the "magical" performance milestone of 1 Peta-flop/s ($10^{15}$ floating point operations per second) at the Munich Supercomputing Centre. In this study we simulated seismic wave propagation under the Merapi Volcano, Java

2014 **Best Poster Award & Best Visualisation Award**, Platform for Advanced Scientific Computing (PASC) Meeting, Zurich, Switzerland

2014 **Best Poster Award**, Volkswagen Foundation Extreme Events, Hannover, Germany

2012 **Outstanding Student Paper Award**, AGU Fall Meeting 2012, San Francisco, USA

2012, 2013 **Travel Grants**, European Center for Geodynamics and Seismology (ECGS) Workshop on Earthquake Source Physics on Various Scales, Luxembourg & Ettore Majorana Foundation and Centre for Scientific Culture (EMFCSC) Workshop on Properties and Processes of Crustal Fault Zones Erice, Italy

2008 **Wilhelm and Else Heraeus Foundation Travel Grant**, for presenting my MSc thesis at the German Physical Society Spring Meeting

2007 **ELISA Scholarship**, Excellency of Women in Science Promotion Program, mentoring at AMD Mask Technology Center Dresden

2006 **DAAD Scholarship**, German Academic Exchange Service fully funded two terms studying abroad at University of Auckland, New Zealand

1995–2002 **Successful Competition**, "Olympiads" of Mathematics, Physics, Informatics, English, Russian

### Awards and Fellowships of Group Members

2026 Croucher Fellowship for Postdoctoral Research, Croucher Foundation Hong Kong, held at Université Côte d'Azur, Nice, France, for PhD student Jeremy Wong

2026 UTIG Distinguished Postdoctoral Fellowship, University of Texas at Austin, for PhD student Jeena Yun

2026 Seismological Laboratory Director's Postdoctoral Scholar, Caltech, for PhD student John M. Rekoske

2026 Cecil and Ida Green Scholar, Cecil H. and Ida M. Green Foundation, for postdoctoral scholar Binhao Wang

2025 NASA FINESST Graduate Student Research Award (3-year support as a Future Investigator in NASA Earth and Space Science and Technology) for PhD student Jonatan Glehman (PI Y. Bock)

2025 CRESCENT Geoscience Professional Development Fellowship, for postdoctoral scholar Bar Oryan

2025 Cecil and Ida Green Scholar, Cecil H. and Ida M. Green Foundation, for postdoctoral scholar Evan Marschall

2025 SSA Student Presentation Award for PhD student Jeena Yun

2025 SSA Paul Andrew Spudich Travel Grant for PhD student Jeena Yun

2025 SSA Paul Andrew Spudich Travel Grant for postdoctoral scholar Baoning Wu

2025 SSA Annual Meeting Student Travel Grant for PhD student Fabian Kutschera

2025 Geo-INQUIRE Transnational Access Award, for Jeremy Wong, 800,000 GPU-hours & exchange in Europe

2025 NHR|WHPC (Women in High Performance Computing), Travel Fellowship for PhD student Iris Christadler

2025 UC San Diego Graduate and Professional Student Association (GPSA) Travel Grants (two) for PhD student Fabian Kutschera, for a summer school in Sicily (8/2025) and the AGU Fall Meeting (12/2025)

2023, 2025 SIO Department Student Awards for Jeena Yun (2023), Jeremy Wong (2025) and John Rekoske (2025)

2024 Cecil and Ida Green Scholar, Cecil H. and Ida M. Green Foundation, for postdoctoral scholar Bar Oryan

2024 John W. Miles Postdoctoral Fellowship in Theoretical and Computational Geophysics, Cecil H. and Ida M. Green Foundation, for postdoctoral scholar Baoning Wu

2024 Eric and Wendy Schmidt AI in Science Postdoctoral Fellowship, UC San Diego, for postdoctoral scholar Yohai Magen

2024 AGU Outstanding Student Presentation Award (OSPA) for PhD student Jeena Yun

2024 NSF-supported NMEM Travel Award, $1,000, for PhD student Fabian Kutschera to attend the Numerical Modeling of Earthquake Motions Workshop in Smolenice Castle, Slovakia

2023 AGU Outstanding Student Presentation Award (OSPA) for PhD student Zihua Niu

2022 Mendenhall Postdoctoral Research Fellowship, USGS, for postdoctoral scholar James Biemiller

2022 Student Joseph Needham Merit Scholarship from the Joseph Needham Foundation for Science & Civilisation in Hong Kong for graduate student Jeremy Wong (UCSD, co-advised with Wenyuan Fan)

2021 SSA Student Presentation Award for MSc student Nico Schliwa (primary adviser)

2017 Best Poster Award, International Workshop on Statistical Modeling (IWSM), 2nd place for A. Bauer (co-adviser)

## Scientific Community Service

### Leadership and Committee Roles in Scientific Organizations

2023–2027 EGU Council and Programme Committee member, EGU Seismology Division President (term ends April 2027): Beno Gutenberg Medal 2026 cycle, chair of the 2027 Seismology Early Career Scientist Award committee evaluation, EGU27 Seismology Division programme

2024– SCEC Science Steering Committee Vice-Chair and Executive Committee member

2024– SCEC Proposal Review Committee (PRC): reviewer in the 2024 cycle (89 proposals), PRC Chair from 11/2024–present, chaired the 2025 and 2026 proposal review cycles (handled 118 and 21 proposals, respectively)

2024–2026 SCEC Annual Meeting science program organization: organized plenary sessions and briefed speakers and moderators (2024, 2025); Annual Meeting Program Committee lead for 2026

2023 SCEC Leadership: Major Activity Leader & Task force: Center-wide Science plan

2021– SCEC Standing Committee for Research Computing and Cyberinfrastructure

2021 SCEC Research Computing and CyberInfrastructure Task Force

2023– SZ4D Steering Committee member; Co-lead, Modeling Collaboratory for Subduction (MCS) Working Group, 2022–2025 Member SZ4D Operations and Planning Committee for Activities (OPC-A)

2022– SZ4D Modeling Collaboratory for Subduction (MCS) Integrative Working Group

2020–2022 SZ4D, Faulting and Earthquake Cycle (FEC) Working Group and Cross Cutting Committee member

2026 TEFRA (Testing Eruption Forecasting through Research and Assimilation) evaluation committee and launch meeting organizing committee (SZ4D; launch meeting 12/2026)

2023– CRESCENT (Cascadia Region Earthquake Science Center) Senior Personnel; co-lead of the DET (Dynamic Rupture, Earthquake Cycle, and Tsunamis) working group

2020– Member Executive Committee, 2021–2023 Chair, CIG – Computational Infrastructure for Geodynamics

2022– Seismic cycles working group, Computational Infrastructure for Geodynamics (CIG)

2026– Co-Chair (with L. Moresi), 2027 Community Meeting of the Computational Infrastructure for Geodynamics (CIG), appointed 04/2026

2020–2022 Steering Committee, MCS-RCN Modeling Collaboratory for Subduction

2020–2024 Scientific Steering Committee, PRACE – European Supercomputer Resources

2020–2022 Seismica (www.seismica.org), Task Force and Editorial Board selection committee

### Award, Prize and Advisory Committees

2026– EarthScope Consortium Innovation and Integration Advisory Committee (IIAC) member (three-year term)

2026– SIAM Activity Group on Geosciences (SIAG/GS) Career Prize and Early Career Prize selection committees

2025– SSA Honors and Awards Encouragement Committee member (three-year term)

2024–2027 AGU William Bowie Medal Committee, member

03/2021–07/2022 SIAM Activity Group on Supercomputing Career Prize Committee

### Program Committees and Conference Organization

2026–2027 Deputy Research Papers Chair, ISC High Performance 2027

2026–2027 Organizing Committee, SIAM Conference on Mathematical and Computational Issues in the Geosciences (GS27)

2026–2027 Discussion Leader, 2027 Subduction Systems Gordon Research Conference, session "Faults and Deformation in Earth's Interior and Surface"

2026 SC26 Technical Papers Program Committee, Applications track, IEEE/ACM International Conference for High Performance Computing, Networking, Storage, and Analysis (paper reviews and virtual committee meeting, 04–06/2026)

06/2026 Co-chair (with X. Yang, Purdue), Technical Session 2 "Advances in Computational Methods and Applications", NSF National Geophysical Facility Community Science Conference, virtual

2025 SC25 Technical Papers Program Committee, Applications track, IEEE/ACM International Conference for High Performance Computing, Networking, Storage, and Analysis

06/2024 Workshop Co-Chair and Scientific Committee, NMEM "Numerical Modeling of Earthquake Motions: Waves and Ruptures" workshop, Smolenice, Slovakia

06/2022 Programme Committee, Galileo Conference "Solid Earth and Geohazards in the Exascale Era"

2021 AGU Fall Meeting Program Committee, Seismology

2020 International Advisory Committee, "International Conference on Computational Engineering and Science for Safety and Environmental Problems (COMPSAFE2020)", Kobe, Japan COMPSAFE2020

2019, 2021 Domain Co-Chair Minisymposia and Posters Program Committee for Solid Earth Dynamics, and Session Convener/Chair, Platform for Advanced Scientific Computing (PASC19, PASC21) Conferences, Switzerland for sessions "Advances in Computational Seismology" and "ACM Papers"

2019 Scientific Committee, "Numerical Modeling of Earthquake Motions: Waves and Ruptures" workshop, Smolenice, Slovakia

### Workshops, Schools and Training Events Organized

2026–2027 Co-organizer, 44th International School of Geophysics "Faults, fluids and the seismic cycle", Erice, Italy (with P. Baccheschi, G. Di Toro and S. Nielsen; school June 27–July 1, 2027)

06/2026 Co-organizer and host, SCEC Tandem Training Workshop "Modeling the Earthquake Cycle from Faults to Supercomputers", SIO/UC San Diego (SCEC award #26020)

06/2026 Co-organizer, Joint SeisSol and ExaHyPE User Workshop, Training and Hackathon "20 Years of SeisSol", Garching, Germany

05/2026 Workshop Organizing Committee, CRESCENT DET Topical Workshop, Eugene, OR, USA

01/2026 Workshop Organizing Committee, CRESCENT Ground Motion Modeling Workshop, virtual and Eugene, OR, USA

07/2025 Co-Organizer, 2nd Tandem hackathon, SIO/UC San Diego

01/2025 Co-organizer, Quakeworx CyberTraining Kickoff Workshop, virtual

11/2024 Co-Organizer, 1st Tandem hackathon, SIO/UC San Diego

10/2024 Organizing Committee, 4th Cargèse International School "Earthquakes: nucleation, triggering, rupture and relationships to aseismic processes", Corsica, France

10/2022 Organizing committee, CIG Seismic Cycles Workshop (https://geodynamics.org/events/details/276)

### Conference Sessions Convened

12/2026 AGU Fall Meeting Session Convener/Chair for 3 sessions (lead convener S007 on data-constrained earthquake source modeling; co-convener S014 on fluids and faulting, and T021 on tools for geodynamic modelling)

05/2026 EGU General Assembly: Seismology Division programme coordination (about 40 sessions); initiator and lead convener of the late-breaking session "Rapid Analyses of Recent Important Earthquakes" (SM0.1); co-convener of sessions SM1.1 and SM2.4; co-convener and chair of the Great Debate GDB2 "The ethics of using Artificial Intelligence in the Geosciences: Opportunities and Risks"

12/2025 AGU Fall Meeting Session Convener/Chair for 5 sessions (lead convener NH003 on digital twins; co-convener S014, T023, T009, and the complex fault systems session)

04/2025 EGU General Assembly: Seismology Division programme coordination; lead convener of sessions SM1.1 and SM2.1; co-convener TS3.4; co-convener of the Great Debate GDB3 "The Role of Large Language Models (LLMs) in Transforming Geosciences: Friend, Foe, or Fad?"; convener of the SM Division Award Lectures

04/2025 SSA Annual Meeting, Baltimore, MD, USA, co-convener, "Numerical Modeling in Seismology"

12/2024 AGU Fall Meeting Session Convener/Chair, "Structure, Mechanics, and Hazards of Geometrically Complex Fault Systems" (T008)

04/2024 EGU General Assembly: Seismology Division programme coordination (Division President); lead convener of sessions SM1.1 and SM2.1; co-convener SM4.17 and TS3.1; co-organizer of the late-breaking session on the 2024 Noto earthquakes; partner (EGU Seismology Division) for the Great Debate GDB2 "Does climate change affect earthquake risks?"

04/2024 SSA Annual Meeting, Anchorage, AK, USA, co-convener of 2 sessions: "Leveraging Cutting-Edge Cyberinfrastructure for Large Scale Data Analysis and Education" and "Illuminating Complex, Multiplet Earthquake Sequences at Kahramanmaraş (Türkiye), Herat (Afghanistan) and Beyond"

12/2023 AGU Fall Meeting Session Convener/Chair, "Data-Informed Earthquake Modeling" (S008)

07/2023 SIAM-GS Minisymposium Organizer for "A deep dive into shallow water"

05/2023 EGU General Assembly Session Convener/Chair for 6 sessions "Earthquake Source Physics: Applications of Imaging, Numerical Simulations and Machine Learning", "The destructive earthquake sequence of February 06, 2023, in south-central Türkiye", "Across the time scales, from earthquakes to earthquake cycle", "Physics-based earthquake modeling and engineering", "General Contributions on Earthquakes, Earth Structure, Seismology", "Division meeting for Seismology (SM)"

12/2022 AGU Fall Meeting Session Convener/Chair for session "Complexities of Fault Rupture and Their Implications for the Joint Studies of Earthquake Source Physics and Ground Motion"

05/2022 EGU General Assembly Session Convener/Chair for 4 sessions, "Earthquake Source Processes: Imaging and Numerical Modeling", "Physics-based earthquake modeling and engineering", "General Contributions on Earthquakes, Earth Structure, Seismology", "Beno Gutenberg Medal Lecture by Yehuda Ben-Zion & SM Division Outstanding ECS Award Lecture by Caroline M. Eakin"

2021 AGU Fall Meeting Session Convener/Chair for 29 sessions, "State-of-the-Art Observations and Modeling of Earthquake Source Processes", "Mechanical Complexity and Structural Heterogeneity in Diffuse Fault Zones: Observation, Modeling, and Experiments", "Unraveling the Complex Interplay Between Short-Term and Long-Term Deformation to Understand Crustal Dynamics", "Seismology Contributions: Earthquake Ground Motion and Engineering Seismology", "Seismology Contributions: Earthquakes", "Seismology Contributions: General", "Seismology Contributions: Structural Seismology"

2020, 2021 EGU General Assembly Session Convener & Chair, General Assembly, Vienna, Austria, for sessions "Physics-based earthquake modeling and engineering", "Earthquake source processes – Imaging methods, physical rupture models and scaling"

2019, 2021 Session Convener/Chair, SIAM GS Conference, mini-symposia "Physics-Based Earthquake and Tsunami Simulation"

2019, 2020 Session Convener/Chair, AGU Fall Meeting, "Advances in Computational Geosciences", "Physics Based Modeling of the Earthquake Cycle. Do Observations Matter?"

2013–2019 Session Convener & Chair, EGU General Assembly, Vienna, Austria, for sessions "Earthquake source processes – Imaging methods, physical rupture models and scaling" and "Numerical modelling of earthquake physics"

2011, 2014 Session Convener/Chair, AGU Fall Meeting, San Francisco, USA, for sessions "Multiscale Dynamics of Earthquake Faulting", "Fault Zone Properties and Processes during Dynamic Rupture"

### Editorial Service

2018– Associate Editor, Journal of Geophysical Research – Solid Earth

2022– Handling Editor and Founding Member, Seismica

2023– Co-led the Seismica special issue on the Cascadia subduction zone with Christie Rowe

2026– Guest Editor of the Geophysical Research Letters special collection "Fault Zone Geometry and Complexity" with P. Martínez-Garzón, Y. Huang and C. Cattania

2026– Co-organizer of the SZ4D AGU special collection (JGR: Solid Earth, with Tectonics, Geophysical Research Letters, G-Cubed and Earth and Space Science)

2021 Guest Editor, Frontiers special issue "High-Performance Computing in Solid Earth Geohazards: Progresses, Achievements and Challenges for a Safer World", all women editors

### Review Panels and Proposal Review

08/2025 U.S. Geological Survey Earthquake Hazards Program, Earthquake Rupture Forecast external grants, 1-day panel

03/2025 NSF EAR Geoinformatics "Sustained Resources" panel, 3-day panel

2025 European Geosciences Union, reviewer for a Galileo Conference proposal (Topical Events Committee)

10/2024 NSF EAR Geophysics panel, 4-day panel

2024–2026 Statewide California Earthquake Center (SCEC): Chair of the Proposal Review Committee for the 2025 cycle (118 proposals) and the 2026 cycle (7 workshop and 14 technical activity group proposals); reviewer in the 2024 cycle (89-proposal cycle)

### Peer Review

Journals, 2012– AGU Advances, Annals of Geophysics, Bulletin of the Seismological Society of America, Computers & Geosciences, Earth and Planetary Science Letters, Earth and Space Science, Geochemistry Geophysics Geosystems, Geology, Geophysical Journal International, Geophysical Research Letters, Journal of Geophysical Research, Journal of Seismology, Mathematics, Philosophical Transactions of the Royal Society A, Nature, Nature Geoscience, Nature Communications, Communications Earth & Environment, International Journal for Numerical and Analytical Methods in Geomechanics, Proceedings of the National Academy of Sciences, Pure and Applied Geophysics, Scientific Reports, Science, Science Advances, Seismica, Seismological Research Letters, Solid Earth, Tectonophysics

Proposals KAUST CRG, French National Research Agency ANR, National Science Foundation (NSF), NERC, ETH Zurich, Earth Observatory Singapore, Barcelona Supercomputing Centre, U.S. Geological Survey, German Research Foundation (DFG), Statewide California Earthquake Center (SCEC)

### External Thesis Examinations

2026 Fatme Ramadan (external examiner, DPhil viva and examiners' report, University of Oxford, UK)

2025 Marvin Roßbach (external examiner and written evaluation of the MSc thesis, Ruhr-Universität Bochum, Germany)

2024 Sebastian Wolf (external examiner and committee member, TUM, Germany)
2024 Lukas Krenz (external examiner and committee member, TUM, Germany)
2022 Magdalena Mittermeier (committee member, LMU Munich)
2022 Markus Schmid (committee member, LMU Munich)
2021 LMU Munich, 1 times committee chair and 3 times committee member
2019 ENS Paris, committee member for Thomas Chartier

### External Hiring Committees and Promotion Evaluations

2025 External evaluations for two promotion and tenure cases
2022 Hiring committee for Tenure track position on AI applied to seismic and /or volcanic hazard, IPGP Paris, France
2022 Hiring committee for Assistant Professor in Seismology, LMU Munich, Germany

## University Service

### UC San Diego and Scripps Institution of Oceanography

2026–2029 SDSU-UC San Diego Joint Doctoral Program (JDP) in Geophysics, Co-Director
2026–2027 SIO Committee on Academic Personnel (CAP), member, Vice Chair in Fall 2026
2023–2026 SIO Committee on Academic Personnel (CAP), member, Acting Chair for two meetings in 1/2026
2023–2026 Academic Senate, Committee on Academic Information Technology (CAIT), member
2026 SIO Strategic Plan implementation team, Goal 1 "Science", member (invited by Director M. Wadhwa)
2026 Participant, Scripps LEADS, the inaugural SIO early- and mid-career faculty professional development program
12/2024–1/2025 Schmidt AI in Science Postdoctoral Fellowship program (campus-wide), reviewer of the SIO applications for the 2025 cohort (with Y. Yu, H. Fricker and I. Eisenman)
2023–2024 Ad hoc academic personnel review committees (2), member

### Institute of Geophysics and Planetary Physics (IGPP)

2020– Green Foundation Selection Committee (including Green Scholar and Miles Fellowship postdoctoral competitions if not conflicted and internal research-support proposals), member
2023– Geophysics Seminar series, faculty co-coordinator (with C. Constable)
2025– Piñon Flat Observatory (PFO) Steering Committee, reconstituted 3/2025 (chair A. Borsa, with D. Agnew, J. Gee, J. Haase), member
2020–2023 Piñon Flat Observatory (PFO) Steering Committee (chair R. Mellors)
2023–2024 NetOps review committee (chair R. Parnell-Turner, with J. Haase and D. Sandwell)

### SIO Curricular Groups

2026–2027 GP Admissions Committee, chair (with D. Stegman and D. May)
2026– Applied Ocean Science (AOS) curricular group, member
2023–2026 GP Departmental Exam Committee, member, 2023–2025; chair, 2025–2026
2024– AOS Departmental Exam Committee, member
2024– GP Geodesy Track Committee (chaired by Y. Bock), member
2022–2023 GP Admissions Committee, co-chair
2020 GP Graduate Program Curriculum Review Committee, member

### LMU Munich

2014–2021 Various faculty committee services at LMU Munich including: redesign of the International Master's program; IT infrastructure; relocation of the Institute into a new building; conference & workshop organisation, such as the ASCETE workshops on earthquake-tsunami modeling in 2014 and 2018 and the German Geophysical Association (DGG) meeting 2020/2022 at LMU

## Research Funding

### Current Grants and Contracts

2026–2029 **EARTH-AID: Agentic AI for Trustworthy Impact Simulations Integrating Digital Twins and Data Lakes for Geohazards & Georesources within Destination Earth** (Horizon Europe, coordinator S. Wiemer, ETH Zurich; 19 beneficiaries), *Horizon Europe*, 9,999,102 € (total); 197,663 € (LMU).
PI of the LMU participation (with H. Igel), funded 02/2026. 2.0% (197,663 €) LMU Munich

2026–2028 **CAIG: Space-time completeness of seismic ground motions via non-intrusive model order reduction** (NSF-RISE-2531036), *National Science Foundation (NSF)*, $1,097,892 (total); $462,400 (SIO share).
Co-PI, PI D. May (UCSD), senior personnel D. Kilb (UCSD). 42.1% ($462,400) SIO; 35.2% ($386,220) UCSD MAE (PI B. Kramer); 22.7% ($249,272) University of Nevada, Reno (PI D. Trugman)

6/2026–5/2027 **Hybrid machine-learning and physics-based ground motion modeling for on-demand earthquake early warning** (USGS-G26AP00141), *U.S. Geological Survey, Earthquake Hazards Program*, $111,335.
PI, with co-Is D. May and D. Kilb (UCSD). 100% ($111,335) UCSD

2026–2027 **CRESCENT 2026–27 Seed Grant: CFM-based splay-fault controls on Cascadia tsunami generation in fully-coupled dynamic rupture-tsunami models**, *CRESCENT (NSF, via University of Oregon)*, $30,000.
PI, proposal developed with PhD student F. Kutschera. 100% ($30,000) UCSD

2025–2028 **MOSAIC: A modular, parallel, 3D seismo-thermo-mechanical approach to simulate the feedback between geodynamic and seismic processes** (DFG Priority Programme SPP 2497 DEFORM "Plate Deformation and Geohazards: The Eastern Margin of the Adriatic Plate", project GA 2465/9-1), *Deutsche Forschungsgemeinschaft (DFG)*, 55,144 € (LMU).
Co-PI, project lead B. Kaus (JGU Mainz), with T. Duretz (Goethe University Frankfurt); PI of the LMU part

2/2025–1/2027 **Construction for the Leadership-Class Computing Facility, Phase 2: Characteristic Science Applications (continuation of the LCCF CSA award)**, *Texas Advanced Computing Center (TACC) through the NSF LCCF award, subaward to UCSD*, $300,000.
PI and project lead. 100% UCSD

2024–2026 **SCEC science awards #24103 (broadband ground motion group modeling), #24127 (CyberTraining for Seismology workshop), #25259 (OneSciencePlace cyberinfrastructure), #25313 (physics-based modeling of postseismic slip in the Ridgecrest fault system using Tandem), #25341 (hierarchical MCMC and dynamic rupture simulations to infer off-fault damage parameters), #26020 (Tandem training workshop)**, *Statewide California Earthquake Center (SCEC)*, 2024: $10,000 (#24103, UCSD share of $35,000), $23,993 (#24127); 2025: $33,817 (#25259), $24,241 (#25313), $25,572 (#25341); 2026: $20,000 (#26020).
PI on #25313, #25341 and #25259 (PI of #25259 after A. Chourasia moved to UCLA); Co-PI on #24127 (lead P. Maechling, USC; award administered at UCSD); Co-I on #24103 (project lead K. Withers, USGS); co-PI on #26020 (PostDoc PI B. Oryan). 100% UCSD, except #24103: 28.6% ($10,000) UCSD, 35.7% ($12,500) UT Austin (D. Liu), 35.7% ($12,500) Caltech (D. Asimaki)

2023–2028 **Quakeworx: An extensible software framework for earthquake simulations (CSSI Frameworks)** (NSF-OAC-2311208 / OAC-2603524), *National Science Foundation (NSF)*, $1,480,000 (UCSD share); $2,500,000 (total).
Co-PI (2023–2025), UCSD lead PI since 10/2025 (OAC-2603524). 59.2% ($1,480,000) UCSD; 22.8% ($570,000) USC (PI Y. Ben-Zion); 18.0% ($450,000) UIUC (PI A. Elbanna)

2023–2028 **CRESCENT: Cascadia Region Earthquake Science Center** (NSF-EAR-2225286), *National Science Foundation (NSF)*, $14,484,604 (center).
Senior Personnel and co-lead of the DET (Dynamic Rupture, Earthquake Cycle, and Tsunamis) working group. Center PI D. Melgar (U Oregon); UCSD subaward 1.7% ($249,999 over 2023–2028; $106,822 received 2023–25 through Year 3)

2023–2027 **TREAD: Data and processes in seismic hazard, Marie Sklodowska-Curie Innovative Training Network (ITN) (Horizon Europe-101072699)**, *Horizon Europe*, 2,664,380 €.
PI, project lead Bruno Pace (Ud'A, Italy). 10% (260,539 €) LMU Munich

2023–2026 **Centre of Excellence for Exascale in Solid Earth (ChEESE-2P)** (EuroHPC-101093038), *Euro HPC & BMBF Germany*, 7,690,590 €.
PI, project lead Arnau Folch (CSIC, Spain). 6% (484,354 €) LMU Munich

2022–2026 **Inferring the source processes of earthquakes from their acoustic-gravity wave impacts on the atmosphere** (NASA- 80NSSC22K0507), *NASA Earth Science Division*, $476,022.
Co-PI, project lead P. Inchin (ERAU). 27% ($129,947) for UCSD. Subaward to UCSD

2022–2026 **PhysMmax: Maximum induced earthquake magnitude in the Groningen Gas Field**, *DeepNL*, 358,228 €.
Co-PI. 7% (25,790 €) LMU Munich

2022–2026 **Geo-INQUIRE: Geosphere INfrastructures for QUestions into Integrated REsearch** (Horizon Europe-101058518), *Horizon Europe*, 14,247,602 €.
PI, project lead Fabrice Cotton (GFZ Potsdam, Germany). 3% (409,432 €) LMU Munich

2021–2026 **FRES Collaborative Research: Toward an integrated modelling framework for physics-based estimates of megathrust rupture potential** (NSF-EAR-2121568), *National Science Foundation (NSF)*, $2,800,000.
PI of the UCSD award (co-PI D. May); collaborative project lead T. Becker (UT Austin). UCSD total $698,099, including a 2024 supplement of $186,136

Completed Grants and Contracts

5/2024–4/2025 **CRESCENT 2024 Seed Grant: Synergizing megathrust seismo-geodetic coupling and slip models using Optimal Transport and Machine Learning**, *CRESCENT (NSF, via University of Oregon)*, $30,000.
PI, proposal developed with postdoctoral scholar B. Oryan. 100% ($30,000) UCSD

2024–2025 **KONWIHR – "GPU Performance and Feature Enhancement of the Earthquake Cycle Simulation Software Tandem"**, *Bavarian State Ministry of Science and the Arts – Competence Network for Technical, Scientific High Performance Computing in Bavaria*, 50,000 € (LMU).
PI; 100% (€50,000) LMU Munich

2024–2025 **ScalaMIDA: Scalable Multi-Index Delayed Acceptance for Non-Linear Earthquake Modelling (EuroHPC Inno4scale innovation study)**, *EuroHPC Joint Undertaking*, 196,500 € (total); 50,063 € (LMU).
PI of the LMU part; coordinator L. Seelinger (KIT), with M. Bader (TUM). 25.5% (50,063 €) LMU Munich; 49.0% (96,375 €) KIT; 25.5% (50,063 €) TUM

2023–2024 **Generation of Near Source Broadband Ground Motion from Physics-based Dynamic Rupture Simulations: Continuing A Group Modeling Approach towards better Characterizing Seismic Hazard for Engineering Application (SCEC award #23121)**, *Southern California Earthquake Center (SCEC)*, $63,000.
Co-PI, project lead Kyle Withers (USGS). 23% ($14,750) UCSD

2022–2025 **DT-Geo: A Digital Twin for GEOphysical extremes** (Horizon Europe-101058129), *Horizon Europe*, 11,866,848 €.
PI, project lead Ramon Carbonell (CSIC, Spain). 6% (731,850 €) LMU Munich

2022–2024 **CSA Leadership-Class Computing Facility: "Characteristic Science Application"** NSF-OAC-2139536, *Texas Advanced Computing Centre (TACC) through the NSF CSA award*, $148,528.
PI and project lead. 100% UCSD

2022–2023 **CyberTraining for Seismology: Data Science and HPC (SCEC award #22162)**, *Southern California Earthquake Center (SCEC)*, $70,328.
Co-PI and project co-lead. 50% ($35,099) UCSD. 50% ($35,229) Marine Denolle (UW)

2022–2023 **Generation of Broadband Ground Motion from Dynamic Rupture Simulations: A Group Modeling Approach towards better Characterizing Seismic Hazard for Engineering Applications (SCEC award #22043)**, *Southern California Earthquake Center (SCEC)*, $73,000.
Co-PI, project lead Kyle Withers (USGS). $1,500 travel support UCSD

2022–2023 **Collaborative Proposal: Distinguishing the near- and the far-field signatures of earthquake sources in complex fault zones (SCEC award #22135)**, *Southern California Earthquake Center (SCEC)*, $49,628.
Co-PI and project co-lead. 54% ($26,817) UCSD. 46% ($22,811) Ahmed Elbanna (UIUC)

2021–2025 **SPIN: Seismological Parameters and Instrumentation, Marie Sklodowska-Curie Innovative Training Network (ITN)**, *European Horizon 2020*, 4,010,000 € (∼4,579,000$).
Co-PI, PI H. Igel (LMU), project lead C. Hadziioannou (UHH). 12.6% (506,000 €) for LMU

2021 **Collaborative Proposal: A Comparative Investigation of Off-Fault Inelastic Processes: From Plasticity to Multiscale Damage**, *Southern California Earthquake Center (SCEC)*, $49,305.
Co-PI, project lead A. Elbanna (UIUC), SCEC award #21112. 50,8% ($25,038 USD) for UCSD

2020–2024 **Geothermie-Allianz Bayern**, *Bavarian State Ministry for Education and Culture, Science and Art*, 8,461,013 € (∼$9,662,000).
Co-PI, PI J. Wassermann (LMU), project lead M. Keim (TUM). 3,6% (308,000 €) for LMU

2020–2021 **Validation of Broadband Ground Motion from Dynamic Rupture Simulations: towards better characterizing seismic hazard for engineering applications**, *Southern California Earthquake Center (SCEC)*, 2020: $59,896; 2021: $75,924.
Co-PI, project lead K. Withers (USGS) and many other Co-PIs, SCEC awards #20046, #21010. LMU 2020: $2,500 USD travel funds

2019–2024 **TEAR – Truly Extended Earthquake Rupture (ERC Starting Grant)**, *European Research Council (ERC)*, 1,500,000 € (∼$1,710,000).
Sole PI and project lead. 100% for LMU Munich. European Research Council funded project for a paradigm shift in modelling how faults slip

2019–2023 **Centre of Excellence for Exascale in Solid Earth (ChEESE)**, *European Horizon 2020*, 7,700,000 € (∼$8,792,000).
PI, project lead: A. Folch (BSC). 6% (450,500 €) for LMU Munich

2019 **Initiation of an international Research cooperation between Bavaria and Brasilia to promote earthquake research**, *BAYLAT*, 3,800 € (∼$4,300).
PI and project lead. 100% for LMU Munich. Travel funds

2018–2021 **SeisSol as a Community Code for Reproducible Computational Seismology (CoCoReCS)**, *German Research Foundation (DFG)*, 490,000 € (∼$572,000).
PI, project lead M. Bader (TUM). 34% (166,650 €) for LMU Munich

2018–2021 **Bayesian inference of earthquake source parameters: kinematic and dynamic finite fault models (BAIES)**, *German (DFG) and Czech (GAČR) Science Foundation*, 400,000 € (∼$467,000).
PI and project lead. 51.5% (205,650 €) for LMU Munich

2018–2021 **Fracture activation in geo-reservoirs – physics of (induced) earthquakes in complex fault networks (FRAGEN)**, *KAUST Competitive Research Grant (CRG)*, $1,050,000.
PI, project lead: P.M. Mai (KAUST). 40% ($420,000 USD) for LMU Munich

2017–2019 **Interdisciplinary earthquake hazard research in Gulf of Aqaba and Strait of Tiran (GAST)**, *KAUST Competitive Research Grant (CRG)*, $1,226,000.
Co-PI, PI P.M. Mai (KAUST), project lead S. Jónsson (KAUST). 7% ($88,000 USD) for LMU Munich

2016–2019 **KONWIHR – NewWave**, *Competence Network for Technical, Scientific High Performance Computing in Bavaria*, 150,000 € (∼$171,000).
Co-PI, PI and project lead H. Igel (LMU). 100% for LMU Munich

2015–2019 **An Exascale Hyperbolic PDE Engine (ExaHyPE)**, *European Horizon 2020*, 2,795,000 € (∼$3,105,000).
PI and Management team, project lead M. Bader (TUM). 10.5% (293,000 €) for LMU Munich

2014–2019 **Advanced Simulation of Coupled Earthquake and Tsunami Events (ASCETE-2)**, *Volkswagen Foundation*, 844,400 € (∼$1,150,000).
PI, project lead: Jörn Behrens (UHH), PI Ylona van Dinther (ETHZ), PI Michael Bader (TUM). 25% (211,000 €) for LMU Munich

2014–2016 **Computational Earthquake Dynamics in Thick Fault Zones**, *BaCaTec*, 28,000 € (∼$38,000).
PI and project lead. Co-PIs: J.-P. Ampuero, Yihe Huang (Caltech, Stanford University, 0%). 100% for LMU Munich. Funds for travel and IT equipment

2014–2016 **KONWIHR – GeoPF**, *Competence Network for Technical, Scientific High Performance Computing in Bavaria*, 135,000 € (∼$184,000).
Co-PI, PI and project lead: H. Igel (LMU). 100% for LMU Munich

2013–2017 **A Collaborative Project: Rupture Dynamics, Validation of the Numerical Simulation Method (CVWS)**, *Southern California Earthquake Center (SCEC)*, 2013: $95,000; 2014: $115,000; 2015: $115,001; 2016: $95,000; 2017: $120,000.
Co-PI, project lead: R. Harris (USGS) and many other Co-PIs in SCEC awards #13061, #14111, #15091, #16056 and #17075. 0% LMU, no-cost collaborator

2013–2015 **High-performance Computing of Earthquake Dynamics on Natural Fault Structures using SeisSol & SORD: A Comparative Study**, *BaCaTec*, 10,000 € (∼$13,500).
PI and project lead. Co-PI: Geoff Ely, USC (0%). 100% for LMU Munich. Funds for travel and IT equipment

2012 **Near-Source Shaking and Dynamic Rupture in Plastic Media**, *Pacific Gas & Electric*, $60,000.
Co-investigator, project lead and PI: P.M. Mai (ETH). 100% ETH. Funds for my PhD position

Fellowships, Internal Awards, Gifts and Service Support

2026–2027 **Green Foundation for Earth Sciences mini-proposal: group GPU server**, *Cecil H. and Ida M. Green Foundation (IGPP)*, $13,000.
PI, jointly with PI D. May. 100% ($13,000) UCSD

2025–2026 **Philanthropic research gifts**, *Pacific Gas and Electric Company (PG&E)*, $112,000 (2025) and $117,695 (2026).
PI and recipient. 100% ($229,695) UCSD

2024–2026 **Thomas and Evelyn Page Chancellor's Endowed Faculty Fellowship in Geosciences**, *UC San Diego*, $100,000.
Faculty Fellow. 100% ($100,000) UCSD

2024–2026 **SCEC Senior Personnel support for SCEC leadership service (SSC Vice-Chair, Proposal Review Committee Chair, Annual Meeting Committee Chair)** (SCEC awards #24505, #25505 and #26505; USC subaward SCON-00006530), *Statewide California Earthquake Center (SCEC)*, 2024: $25,143 (#24505); 2025: $15,671 (#25505); 2026: $15,671 (#26505).
Senior Personnel. 100% ($56,485) UCSD

2024–2025 **High-performance GPU computing for linking earthquakes with geodynamics (A174789)**, *UC San Diego Academic Senate*, $20,000.
PI. 100% ($20,000) UCSD

2024 & 2025 **Green Foundation support for the Tandem Seismic Cycle and Aseismic Slip Supercomputing Hackathons at IGPP (2024 and 2025)**, *Cecil H. and Ida M. Green Foundation for Earth Sciences (IGPP)*, $2,000 (2024); $3,000 (2025).
Co-PI with postdoctoral scholar B. Oryan (lead) and D. May. 100% ($5,000) UCSD

2014–2015 **GeoTutor**, *LMU Center of Leadership and People Management*, 25,000 € and training courses (∼$34,000).
PI and project lead. 100% for LMU Munich. Teaching improvement project for enhanced acquisition of practical skills in geo-software

### High-Performance Computing Allocations

4/2026–4/2027 **CyberShake for fault rupture and ground motion simulations of potential earthquakes at the Eastern Margin of the Adria Plate (Albania) (NSF ACCESS Discover allocation EES260057)**, *NSF ACCESS (Advanced Cyberinfrastructure Coordination Ecosystem: Services and Support)*, ACCESS Discover allocation: 558,348 ACCESS credits and 3,000 node-hours on Stampede3 (TACC) (equivalent to ∼$34,000).
PI, with Co-PIs C. Abril (LMU), S. Callaghan (USC) and K. Olsen (SDSU). 100% UCSD (HPC access)

02/2025–02/2027 **Computational earthquake physics across scales: from induced earthquakes to megathrust subduction modeling** (extension of GCS project pn49ha), *Gauss Centre for Supercomputing*, 41,384,960 core-hours on SuperMUC-NG Phase 1 and 100,000 GPU-hours on SuperMUC-NG Phase 2 (equivalent to ∼238,000 €).
PI and project lead. 100% LMU/UCSD (HPC access)

2025–2027 **Supercomputing of Earthquakes: Unraveling the Physics and Shaking of the Largest Past and Future Earthquakes (TACC Leadership Resource Allocation, project EAR25002)**, *Texas Advanced Computing Center (TACC), NSF Leadership-Class Computing Facility*, 398,400 node-hours: 382,400 on Frontera and 16,000 on Vista (equivalent to ∼$910,000).
PI and project lead, with Co-PI D. May. 100% UCSD (HPC access)

2025–2026 **EuroHPC Regular Access: extreme-scale earthquake simulations with SeisSol** (EHPC-REG-2025R01-043), *EuroHPC Joint Undertaking*, 70,200 node-hours on supercomputer LUMI-G (equivalent to ∼150,000 €).
Co-I, LMU/TUM SeisSol team (HPC access)

2024–2027 **Multiscale physics for next generation earthquake simulators (TACC project EAR24034, allocation of the LCCF Characteristic Science Application)**, *Texas Advanced Computing Center (TACC)*, 272,000 node-hours: 250,000 on Frontera, 15,000 on Vista, 5,000 on Lonestar6 and 2,000 on Horizon (equivalent to ∼$670,000).
Co-PI, PI W. Zhang (TACC). HPC access

2023–2024 **Multiscale physics for next generation earthquake simulations (TACC project EAR23026)**, *Texas Advanced Computing Center (TACC)*, 214,400 node-hours on Frontera and 3,200 node-hours on Lonestar6 (equivalent to ∼$470,000).
PI and project lead. 100% UCSD (HPC access)

2022–2024 **Earthquake seismology across scales: from induced earthquakes to megathrust subduction rupture modeling**, *Gauss Centre for Supercomputing*, 40,652,500 compute hours on supercomputer SuperMUC-NG.
PI and project lead. 100% LMU/UCSD

2021 **QuakeSounds: Mahti Grand Challenge supercomputing project**, *CSC Finland*, 17,6 million computational billing units.
PI, project lead G. Hillers (University Helsinki). Using the fastest supercomputer in the Nordic countries to study seismo-acoustic effects of geothermal stimulation (EGS) induced earthquakes in Helsinki

2020–2021 **3-D seismic wave propagation and earthquake rupture: New roads for the forward and inverse problem**, *Leibniz Supercomputing Centre (LRZ)*, 29,700,000 CPU hours.
PI and project lead. 100% for LMU (HPC access)

2016–2019 **3-D seismic wave propagation and earthquake rupture: New roads for the forward and inverse problem**, *Leibniz Supercomputing Centre (LRZ)*, 33,300,000 CPU hours.
PI and project lead. 100% for LMU (HPC access)

## Mentoring

### Postdoctoral Scholars, Primary or Joint Primary Adviser (17)

2026– **Binhao Wang** (primary adviser, SIO/UCSD, Green Scholar)

2025– **Evan Marschall** (primary adviser, SIO/UCSD, Green Scholar)

2025–2026 **Gabrielle Hobson** (primary adviser, SIO/UCSD, now postdoctoral scholar at MIT)

2024– **Baoning Wu** (primary adviser, SIO/UCSD, Miles Fellow)

2024– **Yohai Magen** (primary adviser, SIO/UCSD, Eric and Wendy Schmidt AI in Science Fellow)

2024–2026 **Bar Oryan** (primary adviser, SIO/UCSD, Green Scholar, now Assistant Professor at Lehigh University)

2024– **Claudia Abril** (primary adviser, LMU)

2022–2026 **Mathilde Marchandon** (primary adviser, LMU, now at ISTerre, Université Grenoble Alpes)

2022–2024 **Anthony Jourdon** (joint primary adviser with D. May, LMU, now Research Engineer at Géoazur Nice, France)

2021–2022 **James Biemiller** (primary adviser, UCSD, Scripps Green's Fellow, now Adjunct Instructor of Geology, Portland State University, USA)

2020– **Thomas Ulrich** (primary adviser, LMU)

2020–2024 **Casper Pranger** (joint primary adviser with D. May, LMU, now industry)

2020–2021 **Carsten Uphoff** (joint primary adviser with D. May, LMU, now industry, AMD)

2019–2022 **Bo Li** (primary adviser, LMU, now Research Scientist at KAUST, Saudi Arabia)

2018–2023 **Duo Li** (primary adviser, LMU, now Senior Seismologist at GNS, New Zealand)

2016–2019 **Kenneth C. Duru** (primary adviser, LMU, now Faculty at University of Texas, El Paso, USA)

2015–2018 **Elizabeth H. Madden** (primary adviser, LMU, now Faculty at San Jose State University, CA, USA)

### Co-mentored and Visiting Postdoctoral Scholars (8)

2025 **Linus Seelinger** (research co-adviser, independent postdoctoral fellow hosted by S. Krumscheid, KIT, Germany, visited SIO/UCSD in 02/2025, departed to industry later in 2025)

2024– **Yeşim Çubuk-Sabuncu** (visiting postdoctoral scholar and research co-mentee, visited SIO/UCSD April–May 2024, primary adviser Benedikt Halldórsson, Icelandic Meteorological Office)

2022–2023 **Ľubica Valentová** (research co-adviser, primary adviser František Gallovič, Charles University, Prague)

2020–2022 **Hugo Sánchez-Reyes** (research co-adviser, primary advisers O. Scotti, S. Hok, IRSN France)

2020–2021 **Maurizio Tavelli** (research co-adviser, primary adviser M. Dumbser, University of Trento)

2020–2021 **Bruno Giammarinaro** (research co-adviser, primary adviser G. Hillers, University of Helsinki)

2018–2020 **Jagdish C. Vyas** (research co-adviser, primary adviser P.M. Mai, KAUST)

2014–2020 **Stefan Vater** (research co-adviser, primary adviser J. Behrens, Hamburg University)

### PhD Students, Primary or Joint Primary Adviser (16)

2024– **Piyush Karki** (joint primary adviser with M. Bader and D. May, LMU&TUM)

2023– **Fabian Kutschera** (primary adviser, SIO/UCSD GP, advanced to candidacy 2026)

2023– **Rachel Preca Trapani** (primary adviser, LMU, TREAD MSCA doctoral fellow, Visiting Graduate Student at SIO/UCSD, 4–5/2026)

2022– **Jonatan Glehman** (joint primary adviser with Y. Bock, SIO/UCSD GP, advanced to candidacy 2025)

2022– **Iris Christadler** (primary adviser, LMU)

2021–2026 **John Rekoske** (primary adviser, co-adviser D. May, SIO/UCSD GP, graduated 5/2026, now Seismological Laboratory Director's Postdoctoral Scholar at Caltech)

2021–2026 **Jeremy Wing Ching Wong** (joint primary adviser with W. Fan, SIO/UCSD GP, defended 6/2026, accepted Croucher postdoctoral scholar fellowship at GéoAzur, Nice, France)

2021–2026 **Jeena Yun** (joint primary adviser with Y. Fialko, SIO/UCSD GP, defended 7/2026, now UTIG Distinguished Postdoctoral Fellow at UT Austin)

2025 **Nico Schliwa** (primary adviser, LMU, graduated with "summa cum laude" 2025, now postdoctoral scholar at ETHZ, visiting student at SIO/UCSD, 5/2023)

2025 **Zihua Niu** (joint primary adviser with H. Igel, LMU, graduated with "summa cum laude" 2025, now industry, Huawei, Visiting Graduate Student at SIO/UCSD, 11–12/2023)

2025 **Jorge Nicolas Hayek** (joint primary adviser with P. Bunge, LMU, graduated with "summa cum laude", 2025, now PostDoc at GFZ Potsdam)

2025 **Sara Aniko Wirp** (primary adviser, LMU, graduated with "magna cum laude", 2025, now at GEOMAR Kiel)

2023 **Taufiqurrahman** (primary adviser, LMU, graduated with "magna cum laude", 2023, now industry, Ionity)

2020 **Thomas Ulrich** (primary adviser, LMU, CRG projects, graduated with "summa cum laude", 2020)

2019 **Sebastian Anger** (primary adviser, LMU, transitioned to an industry position before completing the PhD, 2019)

2018 **Stephanie Wollherr** (primary adviser, LMU, graduated with "magna cum laude", 2018)

Co-mentored and Visiting PhD Students (22)

current **David Schneller** (co-adviser, primary adviser M. Bader, TUM, Visiting Graduate Student at SIO/UCSD, 10/2026–1/2027)

current **Chelsey Assor** (research co-adviser, SDSU-UCSD Joint Doctoral Program in Geophysics, primary adviser R. Almeida, SDSU)

current **Rahim Habibi** (KIT, Germany, primary adviser E. Gaucher & T. Kohl, co-supervised with my postdoctoral scholar T. Ulrich)

current **Daniel Dreier** (Uni. Padua, Italy, primary adviser G. di Toro, co-supervised with M. Marchandon)

current **Francesco Mosconi** (research co-mentee, Sapienza University of Rome, Italy, primary advisers E. Tinti & M. Cocco, visiting graduate researcher at LMU, October 6–18, 2024, and at SIO/UCSD, October–December 2025)

current **Vincent van der Heiden** (research co-adviser, principal adviser Y. van Dinther, U Utrecht)

2025 **Alexis Gauthier** (visiting graduate student at SIO/UCSD in 2025 from Sorbonne University, Paris, France, primary adviser L. Le Pourhiet)

2024/25 **Lorena Kuratle** (visiting social science graduate student at SIO/UCSD from ETH Zurich, Switzerland, 11/2024–1/2025, primary adviser F. Haslinger)

2026 **Vikas Kurapati** (co-adviser, primary adviser M. Bader, TUM, transitioned to an industry position before completing the PhD, 2026)

2024–2025 **Maximilian Kruse** (research co-adviser, primary adviser S. Krumscheid, KIT, Germany, left to industry in 2025)

2024 **Zoe (H.Z.) Yin** (research co-adviser; primary adviser J. Haase, SIO/UCSD GP; graduated 2024; subsequently USGS Mendenhall Postdoctoral Fellow)

2024 **Sebastian Wolf** (co-advised PhD student, Technical University of Munich, primary adviser M. Bader, visiting graduate researcher at SIO/UCSD, May 8–June 2, 2023, graduated 2024, subsequently transitioned to industry)

2024 **Ravil Dorozhinskii** (co-adviser, primary adviser M. Bader, TUM, graduated 2024, now industry, ARM)

2024 **Lukas Krenz** (co-adviser, primary adviser M. Bader, TUM, graduated 2024, now industry, AMD)

2023 **Kadek H. Palgunadi** (research co-adviser, principal adviser P.M. Mai, KAUST, graduated 2023, now faculty at ITS Indonesia)

2022 **Alexander Bauer** (co-adviser, principal adviser M. Küchenhoff, LMU StatLab, graduated 2022)

2022 **Leonhard Rannabauer** (co-adviser, primary adviser M. Bader, TUM, graduated 2022)

2022 **Lauren Abrahams** (research co-adviser, primary adviser E. Dunham, Stanford University, graduated 2022)

2021 **Simone Chiocchetti** (research co-adviser, primary adviser I. Peshkov, University of Trento, graduated 2021)

2021 **Marlon D. Ramos** (research co-adviser, primary adviser Yihe Huang, University of Michigan, graduated 2021)

2020 **Carsten Uphoff** (co-adviser, primary adviser M. Bader, TUM, graduated 2020)

2019 **Iris van Zelst** (co-adviser, primary adviser Y. van Dinther, ETH Zurich, graduated 2019)

UCSD PhD Thesis Committees (10)

current **Matthew Brandin** (SIO/UCSD GP)
current **Jhardel Dasent** (SIO/UCSD GP)
current **Isabel Reinicke** (SIO/UCSD GP)
current **Erick Zavala Sanchez** (UCSD Structural Engineering)
2026 **Ian Vandevert** (SIO/UCSD GP, graduated 2026, chair W. Fan & P. Shearer)
2026 **Xiaoyu Zou** (SIO/UCSD GP, graduated 2026, chair Y. Fialko)
2026 **Ke Xu** (SDSU-UCSD Joint Doctoral Program, graduated 2026, chair K.B. Olsen)
2025 **Ellis Vavra** (SIO/UCSD GP, graduated 2025, chair Y. Fialko)
2025 **Gabrielle Hobson** (SIO/UCSD GP, graduated 2025, chair D. May)
2024 **Zoe Yin** (SIO/UCSD GP, graduated 2024, chair J. Haase)

MSc Students (20)

2026– **Eyad Aljishi** (joint primary adviser with P. Gerstoft, SIO/UCSD GP)
2025– **Bhavik Chandna** (co-adviser with Y. Bock, UC San Diego Computer Science and Engineering MS, SIO 299 research)
2024–2025 **Selin Bulut** (primary adviser, Master of Science Thesis, LMU)
2022/23 **Fabian Kutschera** (primary adviser, Master of Science Thesis, LMU)
2021/22 **Hongyi Su** (primary adviser, Master of Science Thesis, LMU)
2021 **Amrit Bal** (primary adviser, SIO/UCSD GP, graduated with MSc degree in 2021)
2020 **Nico Schliwa** (primary adviser, Master of Science Thesis, LMU)
2019 **Andrea Perez** (primary adviser, Master of Science Thesis, LMU)
2018 **Magdalena Perleth** (primary adviser, Master of Science Thesis, LMU)
2018 **Yongki Andita Aiman** (primary adviser, Master of Science Thesis, LMU)
2018 **Kilian Gessele** (primary adviser, Master of Science Thesis, LMU)
2018 **Tomy Gunawan** (primary adviser, Master of Science Thesis, LMU)
2018 **Mitchel Williams** (primary adviser, Master of Science Thesis, LMU)
2018 **On Ki Angel Ling** (primary adviser, Master of Science Thesis, LMU)
2017 **Ashim Rijal** (primary adviser, Master of Science Thesis, LMU)
2016 **Mirjam Weingärtner** (primary adviser, Master of Science Thesis, LMU)
2016 **Sujania Talavera Soza** (primary adviser, Master of Science Thesis, LMU)
2016 **Alexander Bauer** (co-adviser, primary MSc. adviser: M. Küchenhoff, LMU)
2014 **Amaryllis Nerger** (primary adviser, Master of Science Thesis, LMU)
2014 **Behnam Tang Eysh** (primary adviser, Master of Science Thesis, LMU)
2019–2022 Mentor to several students in each cohort of the LMU Geophysics MSc program

BSc Theses, Primary Adviser (12)

2022 **Moritz Urban** (Bachelor of Science Thesis, LMU)
2022 **Anna Fröhlich** (Bachelor of Science Thesis, LMU)
2021/22 **David Lehr Campos** (Bachelor of Science Thesis, LMU)
2021 **Gizem Yildirim** (Bachelor of Science Thesis, LMU)
2021 **Sophia Gahr** (Bachelor of Science Thesis, LMU)
2021 **Janina Moser** (Bachelor of Science Thesis, LMU)
2020 **Thomas Obermaier** (Bachelor of Science Thesis, LMU)
2020 **Fabian Kutschera** (Bachelor of Science Thesis, LMU)
2019 **Thiemo Bühler** (primary adviser, Bachelor of Science Thesis, LMU)
2019 **Sandra Forndran** (primary adviser, Bachelor of Science Thesis, LMU)
2019 **Nico Schliwa** (primary adviser, Bachelor of Science Thesis, LMU)
2018 **Fabian Risch** (primary adviser, Bachelor of Science Thesis, LMU)

Co-advised Research and Project Students (11)

2023 **Julie Vernet** (Interdisciplinary Project in an Application Subject, LMU/TUM)

2021 **Maximilian Schmeller** (co-adviser, Bachelor of Computer Science Thesis, TUM)

2019 **Maren Kellerer, Asmik Nalmpatia, Markus Ewert & Felix Langer** (co-adviser, research project with Statistics Department of LMU)

2016 **Almond Stoecker & Tobias Kuehn** (co-adviser, research project with Statistics Department of LMU)

2015 **Verena Biek & Bumsun Park** (co-adviser, research project with Statistics Department of LMU)

2014 **Dimitri Pineav** (co-adviser, research project with TUM)

Undergraduate Research Interns (2)

2026 **Charlie Barker** (summer research intern in the TUM PREP program, co-adviser with P. Karki, D. May, M. Bader and T. Ulrich, Imperial College London)

2023 **Emanuel Murphy** (Summer REU Student in RTX program at UT Austin, co-adviser, Fort Valley State University)

## Teaching

UC San Diego

2024–2026 SIOG 227A Introduction to Seismology (4 units), Winter 2024, 2025 and 2026

2024–2026 SIOC 202B Fundamentals of Wave Physics (4 units), seismics half, taken over from J. Hildebrand and redesigned, co-taught with D. Stramski, Spring 2024, 2025 and 2026

2024–2026 SIOG 200B Geophysics Research Skills: Effective Presentation Techniques (2 units), Spring 2024, 2025 and 2026

2023, 2025 SIOG 227C Advanced Seismology: Earthquakes (4 units), newly developed in Spring 2023, co-taught with W. Fan, Spring 2023 and 2025

2026 SIOG 239 High-Performance Computing, Scientific Computation and Numerical Methods for Geophysics (2 units), new course developed from scratch, co-taught with D. May, Spring 2026

2024–2026 SIOG 239 SeismoChat (1–2 units), seminar initiated by Peter Shearer, Wenyuan Fan, and me in Spring 2023, instructor of record Spring 2024, Winter 2025 and Winter 2026

2024 SIOG 239 Rate and State Friction (1 unit), new seminar, co-taught with Y. Fialko, Spring 2024

LMU Munich

2013–2019 Lecturer, LMU Munich, 5 hours/week on Ph.D., M.Sc and B.Sc. level

- Seismology, required lecture and computer lab for M.Sc. students, 2 hours/week, class size approx. 25
- Geophysical Data Analysis, lecture and computer lab for M.Sc. students, 3 hours/week, class size approx. 25
- New Directions in Seismology – Full Waveform Inversion, Slow and Fast Slip, Numerical Earthquake Modelling, seminar class for Ph.D. and M.Sc. students, 2 hours/week, class size approx. 10
- Earthquakes and Faults, seminar class for Ph.D. and M.Sc. students, 2 hours/week, class size approx. 10
- Modern Seismology (co-taught), lecture for M.Sc. students, 2 hours/week, class size approx. 10
- Applied Geophysics 1 (co-taught), required lecture and exercise for B.Sc. students majoring in Geophysics, 4 hours/week, class size approx. 90
- Special Topics in Seismology, required seminar for Ph.D. and M.Sc. students concentrating in seismology, 2 hours/week, class size approx. 10
- Geophysical Colloquium, required seminar for Ph.D. and M.Sc. students covering broad aspects of geophysics, 3 hours/week, class size approx. 40

2022 P7.2 Seismology (semester-long, 2 hours/week; summer 2022). This lecture is mandatory for all Geophysics MSc. students. I have developed this lecture since 2014

2022 WP2.3 Geophysical Colloquium (semester-long, 2 hours/week, with Heiner Igel, Peter Bunge, and Stuart Gilder)

2021/22 WP 5.1 and WP 5.2 New Methods in Seismology (new course, semester-long, 4 hours/week, co-taught with Heiner Igel)

2021/22 WP15.1 Inverse Problems in Geophysics: A Practical Introduction (semester-long, 2 hours/week, with Heiner Igel)

2021/22 New Directions in Earthquake Seismology (semester-long, 2 hours/week)

2021/22 Geophysics Research: Overview on Methods and Open Questions (2 hours/week, co-taught with Heiner Igel, Peter Bunge, Stuart Gilder)

2021/22 Special Topics in Computational Earthquake Physics (2 hr/week, winter 2021/22–summer 2022)

2014–2016 Coordinator and PI "GeoTutor" open tutorials for the acquisition of practical skills in geo-software bringing together undergraduate, graduate, Ph.D. students and departmental faculty teaching each other

### Short Courses, Schools and Training Workshops

2024–2026 Co-organizer and lecturer at 6 high-performance computing training workshops, hackathons and schools broadening access to supercomputing: SCOPED High-Performance Seismology training (UW Seattle, 2024); Quakeworx CyberTraining Kickoff Workshop (2025, virtual, 65 participants from 16 countries selected from 162 applicants, 92% students and postdocs); Tandem Hackathon (SIO/UCSD, 2024); Second Tandem Hackathon (SIO/UCSD, 2025); SCEC Tandem Training Workshop (SIO/UCSD, 2026, 39 participants from 22 institutions in 8 countries); and the joint SeisSol and ExaHyPE user workshop, training and hackathon (Garching, 2026, 63 participants); all free of charge, see also our related 2026 EGU contribution "Equity, diversity, and inclusion initiatives within Geo-INQUIRE – achievements, challenges and good practices"

2023 Lecturer and Co-organizer (with Marine Denolle, UW) of the High-Performance Seismology (HPS) CyberTraining for Seismology

08/2022 Lecturer and Co-organizer of the MTMOD summer school, UT Austin

07/2022 Lecturer of a short-course on dynamic rupture modeling with SeisSol at the EPFL – ETH Zürich, Switzerland, summer school on the "Mechanics of earthquakes and aseismic slip: linking physics-based models to geophysical observations"

05/2022 Lecturer in Shortcourse "Wave Propagation & Computational Seismology", SPIN ITN Doctoral Training Network

10/2021 Lecturer and Co-organizer of the 2nd ChEESE Advanced Training on HPC for Computational Seismology, virtual and free of charge for about 60 participants with preferred access for underrepresented European EU13 country participants

2021 Selected Lecturer at "3rd Cargese Earthquake Summer School on Nucleation, Triggering, and Relationships With Seismic Processes" in Cargèse, Corsica, France. Hands-on tutorial on "HPC dynamic rupture modeling using SeisSol"

2021 Invited Lecturer at SIAM GS21 Conference. Interactive Mini-Tutorial on High-Performance Computing for Computational Seismology and Earthquake Physics using Jupyter notebooks and docker containers

10/2020 Lecturer and Co-organizer of the 1st ChEESE Advanced Training on HPC for Computational Seismology, virtual and free of charge for about 60 participants with preferred access for underrepresented European EU13 country participants

09/2019 Invited Lecturer at the International Center on Theoretical Physics (ICTP) Advanced Workshop on Earthquake Fault Mechanics: Theory, Simulation and Observations, Trieste, Italy (http://indico.ictp.it/event/8715/), training of 60 participants mainly graduate students from developing countries

2019 Lecturer at the four-day long Workshop on Solving Hyperbolic PDE Systems with ExaHyPE at the Leibniz Supercomputing Centre, Germany

2018/19 Teaching of short-courses "Dynamic Rupture on 3D complex fault systems" and "SeisSol: An ADER-DG code for fracture dynamics", KAUST King Abdullah University of Science and Technology, Saudi Arabia

### Teaching and Mentoring Training

2023 Completed Research Mentor Training with Dr. Kelly Diggs-Andrews: 8-hour mentor development workshop to support successful implementation of validated mentorship education interventions

2023 Completed UCSD Training: "Creating A Culture of Care: Supporting Students in Distress"

### ETH Zurich

2008–2012 Instructor, ETH Zurich, Switzerland. Practical geophysics field classes for undergraduate geophysics students

### School Teaching

2010–2011 Grammar School Teacher, Kantonsschule Zurich-Stadelhofen, Switzerland. Mathematics for senior classes (high-school level)

## Invited Talks

Since 09/2019 (105)

09/2026 Conference "Super-shear Earthquakes on Long Straight Faults", University of Oxford, Oxford, UK, Invited Talk

09/2026 Scripps Research Retreat, SIO/UC San Diego, USA, Invited Talk on the research priority "Informed Resilience to Natural Hazards"

07/2026 WCCM-ECCOMAS 2026 Congress, Munich, Germany, Invited Talk, Minisymposium MS264

07/2026 SIAM Conference on Mathematics of Planet Earth (MPE26), Cleveland, OH, USA, Keynote Lecture, "Earthquakes, Tsunamis, and Supercomputers"

06/2026 ISC High Performance 2026, IXPUG Workshop, Hamburg, Germany, Keynote (virtual)

06/2026 Durham HPC Days, Durham University, Durham, UK, Keynote Lecture

06/2026 Seminar, University College London, London, UK, Invited Seminar Speaker

06/2026 ESCO 2026, 10th European Seminar on Computing, Pilsen, Czech Republic, Keynote Lecture (virtual)

06/2026 SCEC UNREST Workshop, Pismo Beach, CA, USA, Invited Talk

05/2026 GAIA "AI for Earth and Ocean at Scripps" Seminar, SIO/UC San Diego, USA, Invited Seminar

04/2026 Department of Earth and Environmental Sciences and Division of Research and Innovation Seminar, San Diego State University, San Diego, CA, USA, Invited Seminar

03/2026 KITP Program "Soft Earth Geophysics", Kavli Institute for Theoretical Physics, UC Santa Barbara, CA, USA, Invited Talk and one-week Program Participant

02/2026 Department of Geosciences Seminar, Princeton University, Princeton, NJ, USA, Invited Seminar Speaker

02/2026 Moody's Earthquake Model Development Seminar, virtual, Invited industry Seminar

01/2026 CRESCENT Ground Motion Modeling Topical Workshop, virtual, Invited Talk (also Workshop Planning Committee)

01/2026 National Academies of Sciences, Committee on Solid Earth Geophysics (COSEG) meeting "Emerging Quantitative Methods in Geophysics", virtual, Invited Panelist

12/2025 AGU Fall Meeting, New Orleans, LA, USA, Invited Talk, Union Session U33C "Earth Song: Advancing Visualization and Sonification in Geoscience"

10/2025 CRESCENT Annual Meeting, Seattle, WA, USA, Invited Presentation

09/2025 AIDT4ES (Artificial Intelligence and Digital Twins for Earth Systems, USACM), Austin, TX, USA, Invited Plenary Lecture

09/2025 3rd Workshop on Earthquake Physics and Applications of Machine Learning to Tectonic Faulting (ERC TECTONIC), Sapienza University/INGV, Rome, Italy, Invited Talk

08/2025 Joint MTMOD and NNW Workshop, St Arnaud, New Zealand, Invited Talk

08/2025 S.N. Jepson Lectureship in Seismology (invited distinguished lectureship), Wellington, New Zealand: public lecture "Earthquake science at the cutting-edge", Government Buildings, Lambton Quay, and research lecture "Fault-size dependent fracture energy and rupture dynamics of cascading earthquakes", Victoria University of Wellington

07/2025 GNS Science subduction zone workshop, Wellington, New Zealand, Invited Talk

05/2025 Earthquake Country Alliance Southern California Workshop, virtual, Invited Public Presentation on the 2025 M5.2 Julian earthquake and the Elsinore fault (Certificate of Appreciation)

04/2025 EGU General Assembly, Vienna, Austria, Invited Talk

03/2025 Symposium honoring Prof. Domenico Giardini, ETH Zurich, Switzerland, Invited Keynote

01/2025 CRESCENT Annual Meeting, University of Washington, Seattle, WA, USA, DET working group presentation

01/2025 USGS Subduction Zone Science Workshop, University of Washington, Seattle, WA, USA, Invited Lightning Talk

12/2024 AGU Fall Meeting, Washington, DC, USA, Invited Talk, Session T018 "Physics-Based Earthquake Forecasting: Insights from Dynamical Models, Laboratory Experiments, Induced Seismicity, and Geophysical Observations"

12/2024 AGU Fall Meeting, Washington, DC, USA, Invited Talk, Session S028 "Recent Advances in Kinematic and Dynamic Source Modeling"

11/2024 "Curiosity Ignited: Research That Sparks Change" Academic Symposium, UC San Diego, USA, Invited Faculty Speaker (public lecture)

11/2024 Department of Earth, Planetary, and Space Sciences Colloquium, UCLA, Los Angeles, CA, USA, Invited Colloquium Speaker

11/2024 SCEC Dynamic Rupture Group Workshop, virtual, Invited Talk

11/2024 TREAD Training School and EPIS Conference on Induced Seismicity, Lunteren, Netherlands, Keynote Lecture

10/2024 4th Cargèse International School "Earthquakes: nucleation, triggering, rupture and relationships to aseismic processes", IESC Cargèse, Corsica, France, Invited Keynote Lecture (and Organizing Committee)

09/2024 Geo-INQUIRE Workshop, GFZ Potsdam, Germany, Invited Talk

06/2024 NMEM "Numerical Modeling of Earthquake Motions: Waves and Ruptures" Workshop, Smolenice Castle, Slovakia, Invited Presentation (also Scientific Committee and Workshop Co-Chair)

06/2024 Department of Earth and Planetary Sciences Seminar, UC Riverside, Riverside, CA, USA, Invited Seminar Speaker

05/2024 SCOPED High-Performance Seismology Workshop, Seattle, WA, USA, Invited Training Lecture (SeisSol)

05/2024 Princeton Center for Theoretical Science Workshop "Fracture across fields: insights from materials science, biology, and geophysics", Princeton University, Princeton, NJ, USA, Invited Speaker

04/2024 EGU General Assembly, Vienna, Austria, Invited Talk, Session ESSI2.1 "Advancing Joint Research on Grand Challenges: International Scientific Infrastructures, FAIR Data, and Open Science"

04/2024 EGU General Assembly, Vienna, Austria, Invited Talk, Session TS3.4 "Active Tectonics and Geodynamics of the Eastern Mediterranean: Special Emphasis on the 6 February 2023 Kahramanmaraș Earthquake Sequence"

04/2024 CISN Earthquake Early Warning Forum, virtual, Invited Presentation (joint with PhD student J. Rekoske)

03/2024 Department of Earth and Space Sciences Colloquium, University of Washington, Seattle, WA, USA, Invited Colloquium Speaker

03/2024 SIAM Conference on Parallel Processing for Scientific Computing (PP24), Baltimore, MD, USA, Invited Talk, Minisymposium "HPC Algorithms for Inverse Problems and Digital Twins"

03/2024 California Community Earth Models for Seismic Hazard Workshop (USGS/SCEC), virtual, Invited Talk

12/2023 AGU Fall Meeting, San Francisco, CA, USA, James B. Macelwane Medal Lecture, "Earthquake Physics and Supercomputing: Deciphering the complex dynamics of the 2023 Kahramanmaraș, Turkey, earthquake doublet"

12/2023 AGU Fall Meeting, San Francisco, CA, USA, Invited Talk, session "Approaches to Networking Global Data Resources" (Geo-INQUIRE)

11/2023 Department of Earth and Planetary Sciences Seminar, UC Davis, Davis, CA, USA, Invited Seminar Speaker

11/2023 Berkeley Seismological Laboratory Seminar, UC Berkeley, Berkeley, CA, USA, Invited Seminar Speaker

10/2023 SSA Topical Conference "Future Directions: Physics-Based Ground Motion Modeling", Vancouver, Canada, Invited Keynote Speaker and Panelist

09/2023 SCEC Annual Meeting, Palm Springs, CA, USA, Invited Plenary Presentation, "Embracing the complexities of earthquake modeling: Connecting observational data and physics-based models"

09/2023 RuFZO (Rupture and Fault Zone Observatory) Workshop at the SCEC Annual Meeting, Palm Springs, CA, USA, Invited Presentation

06/2023 GeoInquire Online Seminar "Simulation Data Lakes", link to recording

06/2023 SIAM-GS, Invited Talk in Session "Advances in Bayesian Inversion for Geoscience Problems"

06/2023 Australasian Leadership Computing Symposium (ALCS), Canberra, Australia, Invited Plenary Speaker

03/2023 ACES (APEC Cooperation for Earthquake Science) International Workshop, Blenheim, NZ, Invited Speaker

01/2023 USGS Subduction Zone Science Workshop, Invited Speaker

12/2022 Leopoldina German National Academy of Sciences, Invited Plenary Panelist "Symposium Earth System Sciences"

12/2022 ERC TECTONIC-FEAR Seminar on Earthquake Physics, link to recording

11/2022 Houston SZ4D Community Meeting

11/2022 Caltech, "Envisioning the Future of Geophysics. A Celebration of the Centennial of the Seismological Laboratory, Caltech", Invited Plenary Speaker

10/2022 Rupture and Fault Zone Observatory (RuFZO) Virtual workshop "Shaping of the Rupture and Fault Zone Observatory", Invited Speaker

10/2022 CIG Seismic Cycle Modeling Symposium, Invited Speaker

07/2022 EPFL – ETH Zürich, Switzerland, summer school on the "Mechanics of earthquakes and aseismic slip: linking physics-based models to geophysical observations", Invited Speaker

07/2022 Platform for Advanced Scientific Computing (PASC) Conference, Zurich, Switzerland, Invited Speaker

07/2022 60 years Leibniz Supercomputing Center (LRZ). Science Panelist

06/2022 SAGE/GAGE Community Science Workshop, Invited Speaker

05/2022 EGU General Assembly, Invited Speaker

04/2022 Mathēmatica Sōlis et Terrae: an Australian Academy of Science – Elizabeth and Frederick White Research Conference, ANU Canberra, Australia, Invited Plenary Speaker

04/2022 EON-ROSE Workshop on Induced Seismicity, Nanaimo, BC, Canada, invited speaker

04/2022 SSA Annual Meeting, Invited Talk in Session "Multi-scale Dynamics of Complex Earthquake Faulting and Seismic Wave Propagation"

04/2022 SSA Annual Meeting, Invited Plenary Talk & Panelist "Frontiers in Seismology"

03/2022 GFZ Potsdam, Invited Seminar Speaker

02/2022 UC Dublin School of Earth Sciences Seminar (cancelled due to Covid-19)

12/2021 AGU Fall Meeting, Invited Speaker

11/2021, 02/2022 Invited Speaker German SPP Round Tables "Plate Deformation and Geohazards: The Eastern Margin of the Adriatic Plate"

11/2021 Stanford Geophysics Virtual Seminar

10/2021 Summer School "Earthquakes: Nucleation, Triggering, and Relationships With seismic Processes", Cargese, Corsica, France, Invited Speaker

09/2021 Fault2SHA Learning Series

08/2021 Sirius Mathematics Center Conference, Sochi, Russia

07/2021 Platform for Advanced Scientific Computing (PASC) Conference, Zurich, Switzerland

06/2021 ISC High Performance Conference, Distinguished Speaker, Frankfurt, Germany

06/2021 SIAM-GS Mini-Tutorial "Computational Seismology" and 2 invited talks, Milano, Italy

06/2021 The Chinese University of Hong Kong Earth System Science Seminar, Hong Kong

06/2021 University of Padua, Department of Geosciences seminar, Padua, Italy

04/2021 ETH Zurich Civil Engineering department seminar, Zurich, Switzerland

04/2021 SSA Annual Meeting, "Exploring Rupture Dynamics and Seismic Wave Propagation along Complex Fault System" session

03/2021 EuroHPC Summit Week, plenary speaker and panelist, Porto, Portugal

12/2020 AGU Fall Meeting, USA

11/2020 Harvard EPS BiSEPPs seminar, Harvard, USA

11/2020 Caltech Seismo Lab seminar, USA

10/2020 Lamont MG&G/SGT seminar, Columbia University, USA

10/2020 USC Earth Sciences Fall seminar, USA

10/2020 SCEC Dynamic Rupture Ingredients Workshop – Rock Properties, California, USA

10/2020 Computational Infrastructure for Geodynamics (CIG) workshop, UC Davis, USA

10/2020 GeoAzur Mardi Seisme seminar, Sophia-Antipolis, France

05/2020 JpGU – AGU Joint Meeting, "Frontier science on solid Earth with HPC" session

04/2020 Berkeley-IPGP Deformation and Tectonics Online Talk Series

03/2020 (event postponed due to the COVID-19 outbreak) International Conference on Computational Engineering and Science for Safety and Environmental Problems (COMPSAFE), Kobe, Japan

03/2020 (event postponed due to the COVID-19 outbreak) Plenary speaker at the German Geophysical Society (DGG) Congress, Munich, Germany

01/2020 SCEC Dynamic Rupture Ingredients Workshop – Fault Friction, Pomona, California, USA

12/2019 AGU Fall Meeting, San Francisco, California, USA

10/2019 MCS-RCN Megathrust Modeling Workshop, Eugene, USA

09/2019 Plenary speaker at the SCEC Annual meeting, Palm Springs, USA

### Selected Invited Talks, 2011–2019 (21)

2019 Invited talk at seismo-tectonic seminar of the Institut de Physique du globe de Paris (IPGP), France

2018 Invited talk at session "Frontiers in Theoretical and Computational Seismology", AGU Fall Meeting, Washington D.C., USA

2018 Invited talk at Fault2SHA workshop "Advancing Seismic Hazard Assessment: Integrating geological and seismological observations into physics-based rupture simulations", King Abdullah University of Science and Technology (KAUST), Saudi Arabia

2018 Invited talks at UTIG and Department of Geosciences, University of Texas, Austin, Texas, USA, link to recording

2018 Invited keynote at workshop "Modeling earthquake source processes (MESP): from tectonics to dynamic rupture", Caltech, Pasadena, USA

2018 Invited keynote at Platform for Advanced Scientific Computing (PASC) Meeting, Basel, Switzerland, link to recording

2018 Invited talk at Seismology of the Americas Meeting session "Exploring Rupture Dynamics and Seismic Wave Propagation along Complex Fault System", Miami, USA

2018 Invited talk at session "Advances in Computational Seismology", European Geosciences Union (EGU) General Assembly, Vienna, Austria

2017 Invited talk at the BGCE Bavarian Graduate School of Computational Engineering Research Day, Munich, Germany

2017 Invited talk at mini-symposium "Deformation accumulation in seismic faults and networks", Scaling Cascades in Complex Systems Conference, Berlin, Germany

2017 Invited talk at mini-symposium "Advances in Numerical Seismic Wave Propagation of Earthquakes, Cancer, Water Tanks and Oil", Platform for Advanced Scientific Computing (PASC) Meeting, Lugano, Switzerland

2017 Invited talk at session "Computational Seismology: From forward to inverse modelling across the scales", EGU General Assembly, Vienna, Austria

2016 Invited talk "Dynamic rupture scenarios from Sumatra to Iceland – High-resolution earthquake source physics on natural fault systems", AGU Fall Meeting, San Francisco, USA

2016 Invited seminar "Earthquake source dynamics: from theory to high-performance computation", Department of Earth Sciences, University of Oxford, UK

2015 Invited talk "High Performance Computing of Earthquake Dynamic Rupture Scenarios on Natural Fault Zones with SeisSol", SCEC 3D Rupture Dynamics Code Validation Workshop, Pomona, California, USA

2014 Invited seminar "Seismic hazard assessment empowered by high performance computing", NatCat Seminar, Allianz Reinsurance, Munich, Germany

2014 Invited seminar "High-frequency earthquake dynamic rupture scenarios", Geophysical Colloquium ETH Zurich, Switzerland

2013 Invited seminar "Large-scale earthquake dynamic rupture scenarios in complex settings", Géoazur, Sophia-Antipolis, France

2013 Invited seminar "Dynamic Rupture in Plastic Media", Institut de Physique du Globe de Paris, France

2012 Invited seminar "Complexities in Earthquake Source Physics", King Abdullah University of Science and Technology, Thuwal, Saudi Arabia

2011 Invited seminar "Transition and Macroscopic Source Properties of Complex Dynamic Rupture Styles", Karlsruhe Institute of Technology, Germany

## Field Work

2013 Ocean-bottom seismic survey, RHUM-RUM project, Prof. K. Sigloch, Réunion

2011 Seismic array survey, Prof. J.-P. Ampuero, San Jacinto Fault, California, USA

2010 GPS survey, Prof. S. Jonsson, Husavik, Iceland

## Diversity, Equity and Inclusion

2025– EGU Equality, Diversity and Inclusion (EDI) Committee, Member and Council representative; assessment of the 2026 committee member applications; also created the EGU Seismology Division EDI officer role (2026)

2026 Lead convener of the AGU26 session "Data-Constrained Earthquake Source Modeling" with invited speakers selected for balance of career stages and gender

2026 Initiator and lead convener of the highly visible EGU invited-abstracts-only session "Rapid Analyses of Recent Important Earthquakes" with an all-women convener team

2025, 2026 Co-convener of EGU Great Debates on large language models and the ethics of AI in the geosciences (2025, 2026)

06–08/2023 Mentor in the TACC Summer REU program, co-adviser to undergraduate REU student Emanuel Murphy (Fort Valley State University), working on containerization of high-performance computing software, incl. attending a mentoring training

2020–2021 ShEESE, Equality Committee charged to promote gender balance within the ChEESE project (e.g., https://youtu.be/h7aImEqJTvk)

2020– Establishing Seismica (www.seismica.org) as member of the Task Force and Editorial Board selection committees promoting equitable access to seismological research, and since 2022 as a Handling Editor. The Seismica Initiative is a community-led effort to establish a Diamond Open Access (free to read and free to publish) journal for seismology

2015–2019 Gender and Equal Opportunities Representative, elected in the scope of the EU funded Horizon2020 "ExaHyPE" project

## Public Outreach and Science Communication

2025 K–12 Science Outreach, Doyle Elementary School, San Diego, CA. Delivered an interactive introduction to seismology and earthquake science to fifth-grade students at a Title I school supporting equitable access to geoscience education at a culturally and socioeconomically diverse public school. In 2024–2025, 41.8% of Doyle students were classified as socioeconomically disadvantaged and 29.7% as English learners

2024–2025 Public engagement: invited faculty speaker at the inaugural UC San Diego Homecoming alumni symposium "Curiosity Ignited: Research That Sparks Change" (2024); Earthquake Country Alliance Southern California preparedness workshop on the 2025 Julian earthquake (Certificate of Appreciation)

Open science communication and education, with a focus on earthquake physics and high-performance computing, using social media (LinkedIn @InSeismoland, about 4,300 followers, Bluesky @inseismoland, about 3,000 followers, X @InSeismoland, about 4,050 followers)

2008–2012 Public Outreach, FocusTerra Earth Science Museum, ETH Zurich, Switzerland. Guided tours, special seminars and dissemination activities to the public

### Media Coverage (since 09/2019)

2026 "20 Jahre SeisSol" ("20 years of SeisSol") – feature article by the Leibniz Supercomputing Centre (LRZ) on our open-source code SeisSol, in German and English, with an interview (July 2026)

2026 "Predicting Tsunami: Tsunami Forecasting Goes Exascale" – Lawrence Livermore National Laboratory Science & Technology Review June 2026 issue, cover and feature story on the Cascadia tsunami early-warning digital twin [98]

2026 "New Satellite's Observations Close Blind Spot in Tsunami Science" by Alex Fox, Scripps Institution of Oceanography and UC San Diego Today news story on Sepulveda et al., Science, 2026 [108]; with independent coverage in Science's "ScienceAdviser: Seeing a tsunami from the sky", Science & Vie (France), and coordinated releases by SDSU, DTU Space, and PUCV

2025 "Shaking the future: How the SeisSol Code is revolutionising earthquake science" – feature article and video produced by the European Centre of Excellence ChEESE on our open-source code SeisSol, with a companion video on Tandem; both distributed on YouTube and social media

2025 "2025 ACM Gordon Bell Prize Awarded for Breakthrough Research on Real-Time Tsunami Forecasting" by Robert Monroe, Scripps Institution of Oceanography and UC San Diego Today news story on Henneking et al., SC'25 [98]; with independent coverage by KPBS, "UCSD's Alice Gabriel wins supercomputing prize for tsunami prediction work", NBC 7 San Diego, Times of San Diego, La Jolla Light, lajolla.ca (feature interview), UT News, HPCwire, City News Service, and the ACM Bulletin

2025 FOX Weather, live national TV interview on the M8.8 Kamchatka earthquake and Pacific-wide tsunami warnings; FOX5 San Diego, "How the tsunami impacted the San Diego coast"

2025 "Preliminary data: Monday's earthquake stronger than anticipated", FOX5 San Diego TV interview, and "How to prepare for the next earthquake in San Diego", KPBS Midday Edition radio interview, on the M5.2 Julian, California, earthquake

2024 "Climate Change-Triggered Landslide Unleashes a 650-Foot Mega-Tsunami", Scripps Institution of Oceanography news story on Svennevig et al., Science, 2024 [73]; with my interviews for NBC News by Evan Bush, The Washington Post by Kasha Patel (article and Post Reports podcast), CBS 8 San Diego (broadcast), and Eos, "Finding the Frequency of a Fjord" by J. Besl; worldwide independent coverage including BBC News, The Guardian, CNN, Scientific American, and New Scientist

2024 Eos Editors' Highlight "Forecasting Earthquake Ruptures from Slow Slip Evolution" by Marcos Moreno on Li and Gabriel, AGU Advances, 2024 [65]; independent editor commentary in AGU's Eos

2024 Eos Editors' Highlight "How Earthquakes Grow from a Tiny Fracture to a Catastrophic Event" by Satoshi Ide on Palgunadi et al., JGR: Solid Earth, 2024 [66]; independent editor commentary in AGU's Eos

2024 "Next-Generation Earthquake Modelling Highlights Differences in Large and Small Earthquake Dynamics" by Eric Gedenk, Gauss Centre for Supercomputing research highlight on Gabriel et al., Science, 2024 [68]

2024 "Small and Large Earthquakes Don't Play by the Same Rules", Scripps Institution of Oceanography news feature on Gabriel et al., Science, 2024 [68]

2023 "Finding Faults" by Sean Cunningham, Texascale (annual magazine of the Texas Advanced Computing Center), Visualizing Science feature on the 3D visualization of our 2019 Ridgecrest earthquake sequence simulations by TACC visualization experts G. Abram and F. Samsel

2023 "Scripps Oceanography to Help Lead New Earthquake Research Center", Scripps Institution of Oceanography news story (adapted from the University of Oregon) on the launch of the NSF Cascadia Region Earthquake Science Center (CRESCENT)

2023 "The Unexpected Physics Behind Turkey's Devastating 2023 Earthquakes" by Alex Fox, Scripps Institution of Oceanography news feature on Jia et al., Science, 2023 [59]

2023 "A SCIENTIST'S LIFE: ALICE GABRIEL. Seismologist uses supercomputing to break down complexity of earthquakes" by Robert Monroe – SIO

2023 "Where will the next 'big one' be in California? Seismologists name faults of top concern" by Danielle Dawson – FOX5 San Diego

2023 "An interview with our new division leader: Alice-Agnes Gabriel" by the EGU Seismology ECS Team

2023 "'Segment-Jumping' Ridgecrest Earthquakes Explored in New Study" by B. Hook – UC San Diego Today

2022 "Years-Long Collaboration Helps Geophysicists Better Understand Severe Earthquake-Tsunami Risks" by E. Gedenk – Research highlight feature article detailing our recent scientific achievements on HPC resources

2021 "How HPC is Shaking Up Modeling of Mysterious Earthquakes" by O. Peckham – HPC Wire article independently reviewing my distinguished talk during ISC21

2021 "Supercomputer creates over 700,000 years of simulated earthquakes" by L. Milideo – Temblor article interviewing me as external expert on Milner et al., BSSA, 2021

2020 "Alice-Agnes Gabriel Charles F. Richter Early Career Award 2020 Recipient" – Seismological Society of America Honors

2020 "We understand only details about earthquakes" by S. Vieser – a conversation about women in science and the future of supercomputers, interview for the LRZ supercomputing centre

2020 "Dr. Alice-Agnes Gabriel wins the 2020 PRACE Ada Lovelace Award for HPC" – HPC Wire article based on PRACE press release

## Professional Memberships

European Geosciences Union (EGU), American Geophysical Union (AGU), Seismological Society of America (SSA), Society for Industrial and Applied Mathematics (SIAM), Association for Computing Machinery (ACM), German Geophysical Society (DGG)

## Other Activities

2019–2020 Board member of the International Montessori Preschool Association, Munich, Germany

2008–2012 Emergency Duties and Communication, Swiss Seismological Service, ETH Zurich, Switzerland

## Peer-reviewed Publications

*undergraduate/graduate student, $^{+}$post-doctoral scholar (co-)advised in the context of the publication.
≈5350 citations, h-index 40, i10-index 91, Google Scholar profile: FmMjT24AAAAJ, list of current preprints

2026 [123] T. Ulrich$^{+}$, X. Zou*, M. Marchandon$^{+}$, N. Schliwa*, F. Tan*, **A.-A. Gabriel**, W. Fan, P. Shearer, M. Thant, T. Zin Htet Tin, E. Lindsey, Y. Fialko, **"On predictability of slip, rupture geometry, and speed of the 2025 M7.8 Mandalay, Myanmar, earthquake"**, Nature Communications, in press, doi:10.1038/s41467-026-77203-x.

2026 [122] H. Sánchez-Reyes$^{+}$, O. Scotti, S. Hok, **A.-A. Gabriel**, **"When Earthquakes Cross the Gap: Physics-Based Dynamic Modeling of Stepover Jumps in Normal Faults"**, Bulletin of the Seismological Society of America, doi:10.1785/0120260099.

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2026 [116] S.A. Wirp*, F. Kutschera*, T. Ulrich$^{+}$, M. Bader, S. Lorito, **A.-A. Gabriel**, **"Hellenic Arc tsunami generation from Mw 8+ 3-D margin-wide dynamic rupture earthquake scenarios"**, Geophysical Journal International, 246 (3), ggag181, doi:10.1093/gji/ggag181.

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